Progress meeting

Closed-form NTK on S^1 (recap + next steps)

Shreyas Kalvankar

March 4, 2026

TU Delft

Recap (last meeting)

What we derived (continuum kernel on S^1)

- **ReLU arc-cosine kernel (closed form):**

$$K(x,y) = \frac{\|x\|\|y\|}{2\pi}\Big(\sin\theta + (\pi-\theta)\cos\theta\Big).$$

- Restrict to S^1: $K(\phi,\psi) = g(\phi-\psi) \quad \Rightarrow$ **convolution operator**.
- **Fourier diagonalization:** $\cos(k\phi), \sin(k\phi)$ are eigenfunctions.

Key spectrum fact (no bias)

$$\lambda_k = \begin{cases} \frac{1}{8}, & k=1 \\ \frac{2}{\pi^2(k^2-1)^2}, & k \text{ is even} \\ 0, & k \text{ is odd, } k>1 \end{cases}$$

Why this matters (training dynamics)

- odd modes ($k = 3, 5, \dots$) are not learned for the no-bias kernel.
- Clean "spectral bias" story in closed form on S^1.

Plan for today

1. **Full NTK with learnable bias:** closed form on S^1.
2. **Fourier eigenvalues:** all frequencies become learnable + rates.
3. **Residual dynamics:** continuum ODE $\Rightarrow$ mode-wise decay.
4. **Discretization (finite n):** kernel matrix dynamics + simple concentration/error controls.

Model and NTK block decomposition (bias included)

$$f(x) = \frac{1}{\sqrt{m}} \sum_{i=1}^{m} a_i \, \sigma(w_i^\top x + b_i), \qquad a_i \sim \mathcal{N}(0,1), \;\; w_i \sim \mathcal{N}(0, I_2), \;\; b_i(0) = 0 \text{ (learnable)}.$$

Let $u = w^\top x + b$, $v = w^\top y + b$, $\sigma(t) = t_+$, $\sigma'(t) = \mathbf{1}_{\{t>0\}}$.

Three parameter blocks

$$\Theta(x,y) = \underbrace{\mathbb{E}[\sigma(u)\sigma(v)]}_{=:K_0(x,y)} + \underbrace{(x^\top y)\mathbb{E}[\sigma'(u)\sigma'(v)]}_{\text{hidden weights}} + \underbrace{\mathbb{E}[\sigma'(u)\sigma'(v)]}_{\text{bias gradients}}.$$

Equivalently,

$$\boxed{\Theta(x,y) = K_0(x,y) + (x^\top y + 1)\, K_1(x,y), \qquad K_1(x,y) := \mathbb{E}[\sigma'(u)\sigma'(v)]}.$$

Restrict to S^1: closed forms for K_0, K_1, and Θ

Parametrize $x(\phi) = (\cos\phi, \sin\phi)$, $y(\psi) = (\cos\psi, \sin\psi)$. Let $\delta \in [0, \pi]$ be the angle: $x(\phi)^\top y(\psi) = \cos\delta$.

K_1 (a quadrant probability)

$$K_1(\delta) = \mathbb{E}[\mathbf{1}_{\{u>0\}}\mathbf{1}_{\{v>0\}}] = \mathbb{P}(u > 0, v > 0) = \frac{\pi - \delta}{2\pi}.$$

K_0 (arc-cosine kernel)

$$K_0(\delta) = \mathbb{E}[\sigma(u)\sigma(v)] = \frac{1}{2\pi}\Big(\sin\delta + (\pi - \delta)\cos\delta\Big).$$

Full NTK on S^1

$$\boxed{\Theta(\delta) = K_0(\delta) + (1 + \cos\delta)K_1(\delta) = \frac{1}{2\pi}\Big(\sin\delta + 2(\pi - \delta)\cos\delta + (\pi - \delta)\Big).}$$

Convolution operator on S^1 and Fourier eigen-decomposition

Let μ be uniform on S^1 and define the continuum operator

$$(Tf)(\phi) = \int_0^{2\pi} \Theta(\phi - \psi)\, f(\psi)\, \frac{d\psi}{2\pi}.$$

Because Θ depends only on $\phi - \psi$, T is convolution $\Rightarrow$ Fourier modes are eigenfunctions:

$$T\cos(k\phi) = \lambda_k \cos(k\phi), \qquad T\sin(k\phi) = \lambda_k \sin(k\phi), \qquad k \geq 1.$$

Eigenvalues can be computed:

$$\boxed{\lambda_k = \frac{1}{\pi} \int_0^{\pi} \Theta(\delta) \cos(k\delta)\, d\delta, \qquad k \geq 0.}$$

Closed-form eigenvalues (bias included): all modes learnable

Exact spectrum

$$\boxed{\lambda_0 = \frac{1}{4} + \frac{3}{\pi^2}, \qquad \lambda_1 = \frac{1}{4} + \frac{1}{\pi^2},}$$

and for higher frequencies:

$$\boxed{\lambda_k = \begin{cases} \dfrac{1}{\pi^2 k^2}, & k \geq 3 \text{ odd}, \\[2ex] \dfrac{k^2+3}{\pi^2 (k^2-1)^2}, & k \geq 2 \text{ even}. \end{cases}}$$

Main takeaway (vs. no-bias kernel)

- **No annihilation:** $\lambda_k > 0$ for every k once the bias-gradient block is present.

Continuum residual dynamics: mode-wise decay

For squared loss under kernel gradient flow in the continuum operator picture:

$$\partial_t r_t = -T r_t, \qquad r_t := f_t - f^\star.$$

Expanding in Fourier:

$$r_t(\phi) = \sum_{k \geq 0} \big(\hat{r}_{c,k}(t)\cos(k\phi) + \hat{r}_{s,k}(t)\sin(k\phi)\big),$$

each coefficient evolves independently:

$$\boxed{\partial_t \hat{r}_k(t) = -\lambda_k\, \hat{r}_k(t) \quad \Rightarrow \quad \hat{r}_k(t) = e^{-\lambda_k t}\hat{r}_k(0).}$$

Consider only the first K modes

Low-frequency subspace on the circle

$$V_K := \mathrm{span}\{1, \cos(\phi), \sin(\phi), \ldots, \cos(K\phi), \sin(K\phi)\} \subset L^2(S^1, \mu).$$

- Think of V_K as "all signals with bandwidth $\leq K$".
- Let P_K be the orthogonal projection (in $L^2(\mu)$) onto V_K.

Key continuum fact

Because T is convolution, it **preserves** Fourier modes:

$$T(V_K) \subseteq V_K \qquad \Longleftrightarrow \qquad \boxed{P_K T = T P_K}.$$

Continuum: projected dynamics are *closed*

Define the low-frequency residual:

$$r_t^{(\leq K)} := P_K r_t \in V_K.$$

Starting from the continuum dynamics $\partial_t r_t = -Tr_t$, we project:

$$\partial_t r_t^{(\leq K)} = P_K \partial_t r_t = -P_K Tr_t = -TP_K r_t = -Tr_t^{(\leq K)}.$$

Interpretation

- In the continuum, the first K modes evolve **as if higher modes don't exist**.
- Each coefficient still decays as $e^{-\lambda_k t}$ for $k \leq K$.

Finite samples: the operator becomes a random matrix

Draw training angles $\phi_1, \ldots, \phi_n \overset{iid}{\sim} \mathrm{Unif}[0, 2\pi)$ and define

$$K_{ij} := \Theta(\phi_i - \phi_j) \in \mathbb{R}^{n \times n}.$$

Residual vector on the sample:

$$r(t) \in \mathbb{R}^n, \qquad r_i(t) = f_t(\phi_i) - y(\phi_i).$$

Under kernel gradient flow:

$$\boxed{\dot{r}(t) = -K\, r(t)}.$$

Define $A := \frac{1}{n} K$ (empirical operator form). Then

$$\boxed{\dot{r}(t) = -nA\, r(t)}.$$

Why finite samples break "convolution"

Continuum (uniform data on the circle)

$$(Tr)(\phi) = \int \Theta(\phi - \psi)\, r(\psi)\, d\mu(\psi), \qquad d\mu(\psi) = \frac{d\psi}{2\pi}.$$

- **Uniform measure:** shifting $\psi \mapsto \psi + \alpha$ does not change μ.
- **Kernel depends on differences:** $\Theta(\phi - \psi)$.
- **So:** T is a convolution operator $\Rightarrow$ Fourier modes diagonalize it.

Finite n (point cloud on the circle)

$$(Ar)_i = \frac{1}{n} \sum_{j=1}^{n} \Theta(\phi_i - \phi_j)\, r_j \qquad \left(A = \tfrac{1}{n} K\right).$$

- The data distribution becomes **spikes**:

$$\mu_n = \frac{1}{n} \sum_{j=1}^{n} \delta_{\phi_j}.$$

- **Not shift-invariant:** rotating the circle moves the spikes.
- **So:** A is not exactly convolution $\Rightarrow$ Fourier vectors are not exact eigenvectors.

Reintroducing Fourier structure: sampled Fourier features

Even though A is not convolution, we can still evaluate Fourier modes on the sample.

Feature matrix for the first K modes
Let $d = 2K + 1$ and define

$$\Phi \in \mathbb{R}^{n \times d}, \qquad \Phi = \left[\, \mathbf{1},\ c_1,\ s_1,\ \ldots,\ c_K,\ s_K \,\right],$$

where $(c_k)_i = \cos(k\phi_i)$ and $(s_k)_i = \sin(k\phi_i)$.

Low-frequency subspace on the sample

$$\mathcal{S}_K := \mathrm{col}(\Phi) \subseteq \mathbb{R}^n.$$

We will track the **projection** of $r(t)$ onto $\mathcal{S}_K$.

Finite-n projected dynamics: within-subspace + leakage

Define the least-squares projection coordinates:

$$\alpha(t) := \arg\min_{a \in \mathbb{R}^d} \|r(t) - \Phi a\|_2^2 \quad \Rightarrow \quad \alpha(t) = (\Phi^\top \Phi)^{-1} \Phi^\top r(t),$$

so that $r_\parallel(t) := \Phi\alpha(t)$ is the component of $r(t)$ in $\mathcal{S}_K$.

Using $\dot{r} = -nAr$ and $r = \Phi\alpha + r_\perp$:

$$\boxed{\dot{\alpha}(t) = -n(\Phi^\top \Phi)^{-1} \Phi^\top A\Phi\, \alpha(t) \; - \; n(\Phi^\top \Phi)^{-1} \Phi^\top A r_\perp(t).}$$

Interpretation

- **Within-subspace matrix:** $(\Phi^\top \Phi)^{-1} \Phi^\top A\Phi$ (a $d \times d$ object).
- **Leakage term:** high-frequency components can feed into low-frequency coordinates because A is not convolution.

Define the empirical matrices

Let $d = 2K + 1$ and define the population matrices:

$$\Lambda := \mathrm{diag}(\lambda_0, \lambda_1, \lambda_1, \ldots, \lambda_K, \lambda_K), \qquad \Sigma := \mathrm{diag}\Big(1, \tfrac{1}{2}, \tfrac{1}{2}, \ldots, \tfrac{1}{2}\Big).$$

Finite-n diagonal correction (self-term):

$$\lambda_k^{(n)} := \Big(1 - \frac{1}{n}\Big)\lambda_k + \frac{1}{n}\Theta(0), \qquad \Lambda^{(n)} := \mathrm{diag}(\lambda_0^{(n)}, \lambda_1^{(n)}, \lambda_1^{(n)}, \ldots, \lambda_K^{(n)}, \lambda_K^{(n)}).$$

Define

$$\boxed{G := \frac{1}{n}\Phi^\top \Phi \in \mathbb{R}^{d\times d}, \qquad H := \frac{1}{n}\Phi^\top A\Phi \in \mathbb{R}^{d\times d}.}$$

The within-subspace dynamics matrix is

$$(\Phi^\top\Phi)^{-1}\Phi^\top A\Phi = (nG)^{-1}(nH) = \boxed{G^{-1}H}.$$

So if $G^{-1}H \approx \Lambda$, then low-mode coefficients decay at (approximately) the population rates.

What we need to show

1) Sampled Fourier features are (almost) orthogonal

$$G := \frac{1}{n}\Phi^\top \Phi \approx \Sigma \quad \text{(entrywise; union bound over } d^2 \text{ entries).}$$

- Ensures $\Phi^\top \Phi$ is invertible and well-conditioned.

2) Restricted operator is (almost) diagonal with the same eigenvalues

$$H := \frac{1}{n}\Phi^\top A \Phi \approx \Sigma \Lambda^{(n)} \quad \text{(entrywise; union bound over } d^2 \text{ entries).}$$

- Then $G^{-1}H \approx \Sigma^{-1}(\Sigma\Lambda^{(n)}) = \Lambda^{(n)} \approx \Lambda$.

3) Approximate eigen-action / invariance (controls leakage)

$$A\Phi \approx \Phi\Lambda^{(n)} \quad \text{(entrywise; union bound over } nd \text{ entries).}$$

What is $\Lambda^{(n)}$ (finite-n diagonal correction)?

In the empirical operator

$$(Au)_i = \frac{1}{n}\sum_{j=1}^{n} \Theta(\phi_i - \phi_j)\, u_j,$$

the sum includes the **self-term** $j = i$, contributing $\frac{1}{n}\Theta(0)\, u_i$.

- In the continuum integral, a single point has measure 0 (no "self-mass").
- In the discrete sum, the self-term has weight $1/n \Rightarrow$ an $O(1/n)$ shift in the *effective eigenvalue* in expectation.

Define corrected eigenvalues (for modes $k = 0, 1, \ldots, K$)

$$\boxed{\lambda_k^{(n)} := \left(1 - \frac{1}{n}\right)\lambda_k + \frac{1}{n}\Theta(0)} \qquad \Rightarrow \qquad \boxed{\Lambda^{(n)} := \mathrm{diag}\big(\lambda_0^{(n)}, \lambda_1^{(n)}, \lambda_1^{(n)}, \ldots, \lambda_K^{(n)}, \lambda_K^{(n)}\big).}$$

As $n \to \infty$, $\Lambda^{(n)} \to \Lambda$.

Lemma A: Gram matrix concentration around Σ

Let $\Sigma := \mathrm{diag}(1, \frac{1}{2}, \frac{1}{2}, \ldots, \frac{1}{2}) \in \mathbb{R}^{d \times d}$.

(A1) Exact expectation

$$\boxed{\mathbb{E}[G] = \Sigma.}$$

(A2) Entrywise Hoeffding bound

Each entry $G_{pq} = \frac{1}{n} \sum_{i=1}^{n} Z_i$ with $Z_i = \Phi_{ip}\Phi_{iq}$ and $|Z_i| \leq 1$. So for any $\varepsilon > 0$,

$$\boxed{\mathbb{P}\big(|G_{pq} - \Sigma_{pq}| \geq \varepsilon\big) \leq 2\exp(-2n\varepsilon^2).}$$

(A3) Union bound over all d^2 entries

$$\boxed{\mathbb{P}\big(\|G - \Sigma\|_{\max} \geq \varepsilon\big) \leq 2d^2 \exp(-2n\varepsilon^2).}$$

Interpretation: sampled Fourier vectors are close to their population orthogonality relations.

(WIP) Lemma B: compressed operator H concentrates around $\Sigma\Lambda^{(n)}$

Recall $H = \frac{1}{n}\Phi^\top A\Phi$ and $\kappa := \|\Theta\|_\infty$.

(B1) Exact expectation

$$\boxed{\mathbb{E}[H] = \Sigma\Lambda^{(n)}.}$$

(B2) Entrywise McDiarmid bound (bounded differences)

For any $\varepsilon > 0$,

$$\boxed{\mathbb{P}\Big(|H_{pq} - (\Sigma\Lambda^{(n)})_{pq}| \geq \varepsilon\Big) \leq 2\exp\left(-\frac{n\varepsilon^2}{8\kappa^2}\right).}$$

(B3) Union bound over all d^2 entries

$$\boxed{\mathbb{P}\Big(\|H - \Sigma\Lambda^{(n)}\|_{\max} \geq \varepsilon\Big) \leq 2d^2\exp\left(-\frac{n\varepsilon^2}{8\kappa^2}\right).}$$

Interpretation: inside the low-mode span, the empirical operator is close to diagonal with diagonal entries $\approx \lambda_k$ (up to Σ and the tiny $\Lambda^{(n)}$ correction).

Lemma C: approximate eigen-action $A\Phi \approx \Phi\Lambda^{(n)}$

Hoeffding + union bound
For each entry (i, p),

$$(A\Phi - \Phi\Lambda^{(n)})_{i,p} = \frac{1}{n}\sum_{j=1}^{n} \underbrace{\Big(\Theta(\phi_i - \phi_j)\varphi_p(\phi_j) - \mathbb{E}[\Theta(\phi_i - \phi_j)\varphi_p(\phi_j) \mid \phi_i]\Big)}_{=:X_j,\ \mathbb{E}[X_j|\phi_i]=0,\ |X_j|\le 2\kappa}$$

For any $\varepsilon > 0$ and $n \geq 2$,

$$\boxed{\mathbb{P}\big(\|A\Phi - \Phi\Lambda^{(n)}\|_{\max} \geq \varepsilon\big) \leq 2nd\,\exp\left(-\frac{n\varepsilon^2}{4\kappa^2}\right).}$$

Interpretation
Each sampled cosine/sine vector is *almost* an eigenvector of A:

$$A\,c_k \approx \lambda_k^{(n)}\,c_k, \qquad A\,s_k \approx \lambda_k^{(n)}\,s_k.$$

Conclusion

Continuum (ideal)

Convolution operator $\Rightarrow$ Fourier modes are **exact eigenfunctions**:

$$\hat{r}_k(t) = e^{-\lambda_k t}\,\hat{r}_k(0).$$

Finite n (practical)

We project residuals onto the first K sampled Fourier features $\mathcal{S}_K = \mathrm{col}(\Phi)$ and track coefficients $\alpha(t)$.

The projected dynamics are governed by the **restricted operator** $G^{-1}H$:

$$\dot{\alpha}(t) \;=\; -n\,(G^{-1}H)\,\alpha(t) \;+\; \text{(small leakage)}.$$

Main message

With enough samples, the low-mode geometry and restricted operator **concentrate**:

$$G^{-1}H \approx \Lambda \qquad \Longrightarrow \qquad \boxed{\dot{\alpha}_k(t) \approx -n\lambda_k\,\alpha_k(t)} \quad (k \le K).$$

So the first K Fourier modes still decay at (approximately) the **population rates** dictated by λ_k.

Next steps

1) Finish the finite-n story (turn lemmas into a theorem)

- Combine Lemma A/B/C into a single **high-probability** statement:

$$G^{-1}H \approx \Lambda \quad \Rightarrow \quad \dot{\alpha}_k(t) \approx -n\lambda_k\, \alpha_k(t) \ (k \le K),$$

  with an explicit sample-size requirement $n = n(K, \varepsilon, \delta)$.

- Clarify what assumptions are needed to make the **leakage** term small (e.g. low-frequency target / initialization).

2) Empirical validation

- Simulate finite-n kernel GD on S^1: compare $r(t)$ projected onto first K modes vs. the prediction $e^{-n\lambda_k t}$.
- Sweep n and K to see when mode-wise decay becomes stable.

3) Move onto finite-width

- **Finite width:** quantify deviation of Θ_m from Θ (kernel concentration / drift).